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DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

Internet Of Things

Challenge 2 – Exercise

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Scenario recap and assumptions

A battery-powered temperature sensor publishes a reading every 2 minutes (30 cycles per hour). A battery-powered actuator (fan controller) must receive this reading and react accordingly. A mains-powered gateway is available, but it only speaks MQTT. Two options are compared:

- **Option A (CoAP):** sensor → actuator directly, CON PUT + piggybacked ACK.
- **Option B (MQTT via gateway):** sensor publishes to the broker, actuator subscribes to the broker.

Message sizes (header + payload) as provided

CoAP	Size	MQTT	Size
CON PUT Request	70 B	Subscribe	65 B
PUT ACK	15 B	Sub Ack	55 B
		Publish (QoS 1)	75 B
		Pub Ack	50 B
		Connect / Disconnect	60 B
		Connect Ack	50 B

Adopted assumptions

- $E_{\mathrm{TX}} = 50\,\mathrm{nJ\,bit^{-1}}$ and $E_{\mathrm{RX}} = 58\,\mathrm{nJ\,bit^{-1}}$.
- CoAP uses Confirmable (CON) messages; MQTT uses QoS 1 on both sides.
- The channel is lossless (no retransmissions are counted in EQ1).
- Devices start disconnected. **For EQ1** the sensor and the actuator perform a single connection setup at the beginning of the hour and remain connected throughout — the sensor continuously publishes and the actuator continuously receives, so no per-measurement DISCONNECT is issued. This is the natural reading of "no sleep" and it avoids the subtle asymmetry that reconnecting every 2 minutes would introduce with QoS 1.
- Only the communication energy consumed by the two battery-powered endpoints is counted; the gateway / broker is on mains and is ignored.

Energy model

For a transmitted message of B bytes, $E_{\mathrm{TX,msg}} = B \cdot 8 \cdot E_{\mathrm{TX}}$. For a received message of B bytes, $E_{\mathrm{RX,msg}} = B \cdot 8 \cdot E_{\mathrm{RX}}$. Intermediate computations are in nJ; final results are reported in µJ ($1\,\mu\mathrm{J} = 1000\,\mathrm{nJ}$).

EQ1a — CoAP direct communication

Every 2-minute cycle the sensor sends a single CON PUT (70 B = 560 b) to the actuator, which answers with a piggybacked ACK (15 B = 120 b). The sensor transmits the request and receives the ACK; the actuator does the opposite.

Per-cycle energy — sensor (TX 70 B, RX 15 B)

$$\begin{aligned} E_{\text{sensor/cycle}} &= 560 \cdot 50 + 120 \cdot 58 \\ &= 28{,}000 + 6{,}960 = \boxed{34\,960\,\text{nJ}} \end{aligned}$$

Per-cycle energy — actuator (RX 70 B, TX 15 B)

$$\begin{aligned} E_{\text{act/cycle}} &= 560 \cdot 58 + 120 \cdot 50 \\ &= 32{,}480 + 6{,}000 = \boxed{38\,480\,\text{nJ}} \end{aligned}$$

Hourly totals (30 cycles)

$$\begin{aligned} E_{\text{sensor,1h}} &= 30 \cdot 34{,}960 = 1{,}048{,}800\ \text{nJ} = 1048.80\,\mu\text{J} \\ E_{\text{act,1h}} &= 30 \cdot 38{,}480 = 1{,}154{,}400\ \text{nJ} = 1154.40\,\mu\text{J} \\ E_{\text{CoAP,1h}} &= 1048.80 + 1154.40 = \boxed{2203.20\,\mu\text{J}} \end{aligned}$$

Answer: EQ1a = 2203.20 µJ per hour (sensor + actuator combined).

EQ1b — MQTT via the gateway

Under the adopted interpretation (a single setup at the beginning of the hour, then continuous operation), each endpoint pays a one-off setup cost and then a per-measurement cost for every cycle.

Sensor as MQTT publisher

Setup (once): CONNECT (60 B TX) → CONNACK (50 B RX).

$$\begin{aligned} E_{\text{sensor,setup}} &= 60 \cdot 8 \cdot 50 + 50 \cdot 8 \cdot 58 \\ &= 24{,}000 + 23{,}200 = \boxed{47\,200\,\text{nJ}} \end{aligned}$$

Per cycle: PUBLISH QoS 1 (75 B TX) → PUBACK (50 B RX).

$$\begin{aligned} E_{\text{sensor/cycle}} &= 75 \cdot 8 \cdot 50 + 50 \cdot 8 \cdot 58 \\ &= 30{,}000 + 23{,}200 = \boxed{53\,200\,\text{nJ}} \end{aligned}$$

Hourly sensor total (30 cycles):

$$\begin{aligned} E_{\text{sensor,1h}} &= 47{,}200 + 30 \cdot 53{,}200 \\ &= 47{,}200 + 1{,}596{,}000 \\ &= 1{,}643{,}200\ \text{nJ} = \boxed{1643.20\,\mu\text{J}} \end{aligned}$$

Actuator as MQTT subscriber

Setup (once): CONNECT (60 B TX) → CONNACK (50 B RX) → SUBSCRIBE (65 B TX) → SUBACK (55 B RX).

$$\begin{aligned} E_{\text{act,setup}} &= 60 \cdot 8 \cdot 50 + 50 \cdot 8 \cdot 58 \\ &\quad + 65 \cdot 8 \cdot 50 + 55 \cdot 8 \cdot 58 \\ &= 24{,}000 + 23{,}200 + 26{,}000 + 25{,}520 \\ &= \boxed{98\,720\,\text{nJ}} \end{aligned}$$

Per cycle: PUBLISH QoS 1 (75 B RX) → PUBACK (50 B TX).

$$\begin{aligned} E_{\text{act/cycle}} &= 75 \cdot 8 \cdot 58 + 50 \cdot 8 \cdot 50 \\ &= 34{,}800 + 20{,}000 = \boxed{54\,800\,\text{nJ}} \end{aligned}$$

Hourly actuator total (30 cycles):

$$\begin{aligned} E_{\text{act,1h}} &= 98{,}720 + 30 \cdot 54{,}800 \\ &= 98{,}720 + 1{,}644{,}000 \\ &= 1{,}742{,}720\ \text{nJ} = \boxed{1742.72\,\mu\text{J}} \end{aligned}$$

MQTT total (sensor + actuator)

$$E_{\mathrm{MQTT,1h}} = 1643.20 + 1742.72 = \boxed{3385.92\,\mu\mathrm{J}}$$

Answer: EQ1b = 3385.92 µJ per hour (sensor + actuator combined), i.e. roughly 1.54× the CoAP-direct solution.

EQ1 summary

Protocol	Sensor / h	Actuator / h	Total / h
CoAP (direct, CON)	1048.80 µJ	1154.40 µJ	2203.20 µJ
MQTT (via gateway, QoS 1)	1643.20 µJ	1742.72 µJ	3385.92 µJ

Under these assumptions, CoAP consumes less communication energy because it avoids the MQTT connection/subscription overhead and uses a much smaller acknowledgement per measurement (15 B vs 50 B PUBACK, no separate SUBSCRIBE/SUBACK for the actuator).

EQ2 — Actuator wakes up once every 30 minutes

When the actuator is asleep most of the time, the direct-CoAP option degrades sharply. The sensor still publishes every 2 minutes, but every CON PUT now targets an unreachable peer for 28 out of every 30 minutes. Each failed request triggers the full retransmission schedule (RTO = 30 s, MAX_TRANSMISSIONS = 3), wasting sensor energy on up to 3 transmissions per cycle with no hope of delivery, and the actuator still misses everything that was sent while it was offline.

MQTT solves this cleanly. The sensor publishes with **retain = 1**, so the broker always stores the freshest temperature on the topic. The actuator uses a **persistent session** (`cleanSession = false`) with QoS 1: on wake-up it reconnects, the subscription is still active broker-side (no re-SUBSCRIBE needed), the retained message is delivered immediately, and any QoS-1 messages queued while the actuator was offline are flushed in order. After processing the actuator sends PUBACK, DISCONNECTs, and sleeps again.

Actuator wake-up energy budget (steady state)

Each wake-up the actuator performs:

```
CONNECT (60 B TX) -> CONNACK (50 B RX) ->
  (retained/queued) PUBLISH QoS 1 (75 B RX) -> PUBACK (50 B TX) ->
  DISCONNECT (60 B TX, not ACKed)
```

$$
\begin{aligned}
E_{\text{wake}} &= 60 \cdot 8 \cdot 50 + 50 \cdot 8 \cdot 58 + 75 \cdot 8 \cdot 58 + 50 \cdot 8 \cdot 50 + 60 \cdot 8 \cdot 50 \\
&= 24{,}000 + 23{,}200 + 34{,}800 + 20{,}000 + 24{,}000 \\
&= 126{,}000 \text{ nJ} = 126.00\,\mu\text{J}
\end{aligned}
$$

With two wake-ups per hour (at $t = 30$ and $t = 60$ min):

$$E_{\text{act,EQ2,1h}} = 2 \cdot 126.00 = 252.00\,\mu\text{J/h}.$$

For comparison the actuator paid 1742.72 µJ/h under EQ1b (always connected, every cycle), so switching to a sleep-then-retained-fetch pattern reduces the actuator-side energy by roughly a factor of seven while still delivering the freshest reading on every wake-up.

Choice and justification

Answer: Use MQTT with `retain = 1`, `QoS = 1`, `cleanSession = false`. Direct CoAP CON fails every sensor cycle (actuator asleep ⇒ RTO retransmits ⇒ wasted sensor energy, no delivery). Retained + persistent MQTT lets the actuator wake every 30 min, skip re-subscribing, and receive the latest value plus any queued updates. Metric optimised: actuator-side energy per hour and delivery reliability.